Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

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Visualization and Highlighting of Ground-Glass Opacities in Chest CT

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1. Problem Statement

Ground Glass Opacities (GGOs) appear as faint cloudy areas seen in lung CT scans [1]. They can be early signs of diseases such as pneumonia, lung infections, or even cancer. Finding these areas by hand is difficult and consumes a large amount of time because GGOs are low contrast areas and spread out gradually in the lungs. This could result in different radiologists interpreting the same CT scan differently, which may affect diagnosis accuracy. Without a well organized preprocessing pipeline, steps like segmenting the lungs, detecting GGOs, and measuring their size can cause inconsistent results, which makes it hard to perform reliable automated analysis.

2. Objectives

The main objectives of this project are:

- Improve the quality of the original chest CT scans by using various image processing techniques.
- Accurately segment lung regions using HU based thresholding and morphological operations while removing noise.
- Detect Ground Glass Opacity regions within the segmented lung area using HU based thresholding criteria.
- Evaluate the effectiveness of the proposed image processing pipeline using a publicly available chest CT dataset.
- Design a simple prototype application which focuses on clear GGO visualization.

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3. Literature Review

3.1. Study 1: Hounsfield Unit-Based Lung Segmentation

Early automated lung segmentation methods primarily relied on intensity thresholding, exploiting the low radiodensity of air-filled lung parenchyma relative to surrounding thoracic tissues. Attenuation values expressed in Hounsfield units (HU) provide characteristic ranges for different anatomical structures on CT images, enabling basic separation of lung regions from the chest wall and mediastinum [2]. Typically, lung tissue falls within a range of approximately -1000 to -500 HU. While effective for healthy lungs, this approach becomes unreliable in the presence of pathology, where abnormal tissue alters normal intensity distributions.

To address these limitations, fully automated three-dimensional segmentation frameworks were developed that combined thresholding with structural optimization techniques. Hu et al. introduced a method integrating gray-level thresholding, dynamic programming, and morphological operations to refine lung boundaries [3]. Dynamic programming was used to optimize boundary placement in regions with weak intensity gradients, while morphological operations corrected artifacts such as holes and isolated regions. This approach established early performance benchmarks for automated lung segmentation.

More recent practical implementations have focused on usability and reproducibility. Hartmann et al. demonstrated a Python-based workflow using fixed HU thresholds followed by morphological closing applied separately to the left and right lungs [4]. Processing the lungs independently prevented artificial fusion near the mediastinum, where anatomical proximity often confounds binary segmentation. These frameworks emphasize accessibility and rapid prototyping while maintaining anatomical plausibility.

3.2. Study 2: Advanced Segmentation for Diseased Lungs

Segmentation becomes significantly more challenging in diseased lungs due to fibrosis, consolidation, or interstitial abnormalities that disrupt normal intensity patterns. To address this, researchers have combined intensity-based methods with texture analysis. Uppaluri et al. employed gray-level thresholding together with texture-feature images derived from gray-level co-occurrence matrices (GLCMs) to segment lungs affected by severe interstitial disease [5]. Texture features captured spatial relationships beyond simple intensity values, enabling improved separation of lung tissue from surrounding structures. The method achieved approximately 82% overlap with expert manual annotations.

Hybrid approaches integrating soft classification have further improved robustness. Fuzzy c-means clustering allows voxels to belong partially to multiple tissue classes, reflecting gradual biological transitions rather than hard boundaries. One such study combined fuzzy clustering with morphological hole filling, dilation, and reconstruction to specifically address juxtapleural nodules that adhere to the chest wall [6]. Morphological reconstruction selectively preserved anatomically meaningful boundaries while filling incomplete regions. These methods proved effective when disease obscured normal lung-chest wall interfaces.

3.3. Study 3: Ground Glass Opacity Detection Algorithms

After lung segmentation, identifying ground glass opacities (GGOs) requires distinguishing subtle increases in attenuation from normal lung parenchyma. Probabilistic spatial models have been widely adopted for this purpose. Markov random field (MRF)-based frameworks incorporate neighborhood dependencies, modeling the diffuse and spatially coherent nature of GGOs [7]. Using contextual voxel relationships, one such method achieved 86.94% sensitivity and 94.33% specificity. To reduce false positives, vessel structures were removed using morphological filters targeting elongated, line-like geometries.

Statistical characterization methods provide an alternative perspective. In one approach, candidate GGO regions were first segmented using morphological reconstruction, followed by computation of first-order statistical descriptors such as mean, variance, entropy, skewness, and kurtosis [8]. Entropy was particularly effective in capturing the heterogeneous texture typical of GGOs. This framework enabled probabilistic confidence scoring rather than strict binary classification.

Texture-based local pattern analysis has shown strong performance in detecting nodular GGOs. Kim et al. proposed a boosted k-nearest neighbor classifier using nonparametric density estimation, followed by segmentation via texture likelihood maps [9]. The method successfully detected all GGOs in the evaluated dataset with minimal false positives. Integrating detection and segmentation into a unified pipeline improved overall robustness.

Recent clinical demands, particularly during the COVID-19 pandemic, have emphasized computational efficiency. An attention-based thresholding method selectively emphasized diagnostically relevant regions while suppressing background noise, improving Dice similarity by 8.9% while requiring less than 0.1% of the computational cost of deep learning models [10]. Such efficiency enables deployment on standard clinical hardware.

Three-dimensional geometric analysis has introduced new characterization capabilities. PointNet++ architectures applied to volumetric GGO segmentation achieved up to 98% classification accuracy and 92% intersection-over-union [11]. Minkowski tensors were used to quantify shape anisotropy and morphological complexity, enabling differentiation between GGO subtypes based on three-dimensional structure.

3.4. Study 4: Pulmonary Vessel Segmentation and Suppression

Pulmonary vessels pose a major challenge for GGO detection because their attenuation values overlap with those of ground glass regions. Multiscale vessel enhancement filters based on Hessian matrix eigenvalue analysis have become a standard solution. Frangi et al. introduced a vesselness measure that identifies tubular structures using local second-order intensity variations [12]. Subsequent three-dimensional implementations achieved vessel segmentation accuracies between 96.2% and 97.0% [13]. Multiscale analysis enables detection of vessels ranging from central arteries to peripheral branches.

Dynamic contour-based methods have also been applied to vessel extraction. These approaches typically combine threshold segmentation with morphological filtering and geometric constraints, such as elliptical fitting based on vessel cross-sectional shape [14]. Contour tracking preserves connectivity along vessel paths despite local intensity variation.

Vessel segmentation has also been explored for quantitative analysis. Studies focusing on vessel tortuosity applied gray-level thresholding followed by morphological closing and Hessian-based enhancement filters [15]. Distance transform techniques were then used to quantify vessel proximity, providing spatial context that can inform region-specific vessel suppression strategies.

3.5. Gap Analysis

Despite substantial progress, several limitations remain in current GGO detection systems. First, most methods operate as fully automated pipelines with limited user interaction, preventing clinicians from adjusting parameters based on patient-specific anatomy or pathology. Second, vessel suppression techniques often apply uniform parameters across the entire lung volume, ignoring regional variations in vascular density. This can lead to insufficient suppression in central regions or excessive erosion in peripheral areas.

Third, validation studies typically report aggregate performance metrics without detailed analysis of failure modes or edge cases. Understanding where and why algorithms fail is critical for targeted improvement. Fourth, few systems provide volumetric quantification with slice-by-slice navigation, limiting correlation between three-dimensional disease extent and clinical severity. Fifth, visualization features such as adjustable overlay transparency are rarely emphasized, despite their importance for clinical interpretation.

Finally, computational efficiency is often evaluated only at the algorithm level, neglecting the cumulative cost of preprocessing steps such as DICOM loading, windowing, segmentation, and visualization. These gaps motivate the development of interactive visualization tools that support adjustable parameters, region-aware processing, detailed validation metrics, volumetric navigation, transparency control, and end-to-end pipeline optimization. Such tools are essential for bridging the gap between research prototypes and practical clinical deployment.

4. Methodology

The project will consist of 4 main steps:

1. Data Collection and loading
2. Data Preprocessing
3. Lung segmentation
4. GGO visualization and Evaluation

The overall methodology is illustrated in figure 1:

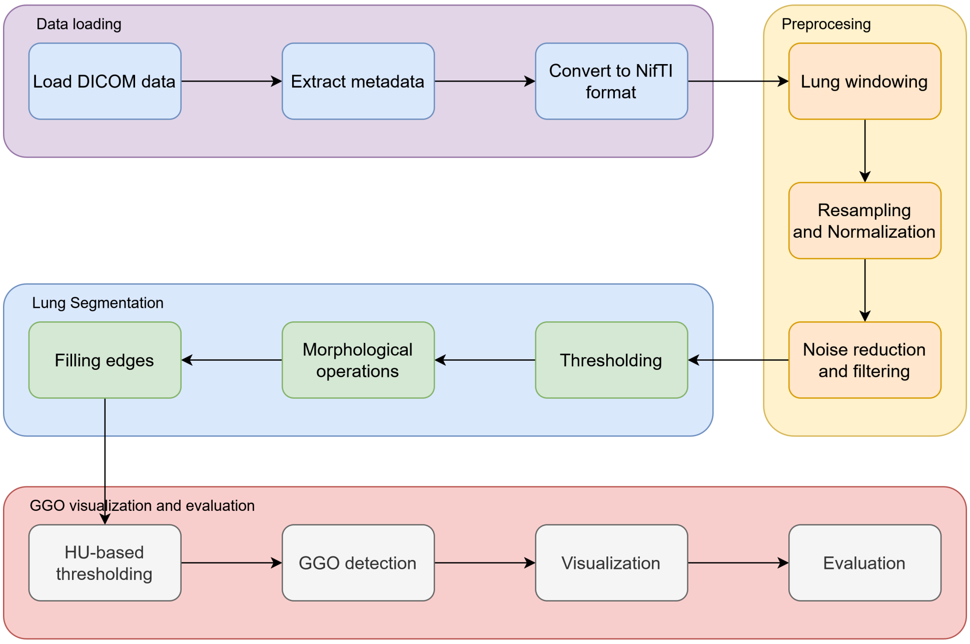

Figure 1: Methodology Overview

4.1. Data Collection and loading

We will use the publically available LIDC-IDRI dataset [16] which contains lung CT scans from the Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI). The dataset contains over 11000 clinical thoracic CT scans in DICOM format. The Python library "pylidc" is the standard tool used to query and visualize this data. The data will be loaded using Python libraries such as "pydicom".

Once the dataset is loaded, we will extract their metadata. The metadata is primarily contained within the DICOM headers of the images and the XML files that hold the radiologist's findings. The LIDC-IDRI dataset provides metadata under three categories:

- **Scan and Patient information:** Includes slice thickness, pixel spacing, intravenous contrast usage

- **Annotations:** Includes type of Lesion and ROI maps
- **Nodule characteristics:** For nodules ¿ 3mm, the metadata includes subjective ratings from 1 to 5 for nine specific features

Once the metadata is extracted, we will convert the DICOM images to standard NifTI format using the "dcm2niix" tool for easier processing in later stages. This conversion helps for compatibility with various medical image processing libraries as DICOM is complex and bulky for analysis.

4.2. Data preprocessing

After loading and extracting metadata, the data will be preprocessed to improve the overall quality of the CT images before segmentation. This step includes:

- Windowing the images to focus on lung tissue
- Resampling and Normalizing pixel intensities
- Removing noise and artifacts

We will apply a lung-specific window level and width to enhance the visibility of lung tissues. This is a medical image convention used to clip the Housenfield range is by choosing a central intensity. Next, resampling will be done to ensure uniform voxel spacing across all scans, which is important for accurate segmentation. This is done because, different scanners may produce images with varying resolutions. Resampling changes the resolution to a standard size such as 1mm x 1mm x 1mm. We will use interpolation methods such as linear or cubic interpolation for this purpose.

Once resampling is done, normalization of pixel intensities will be performed to standardize the intensity values across different scans. This makes mathematical operations more stable and removes dependency on absolute HU values for algorithms. Noise reduction techniques such as Gaussian, Median or Bilateral filtering will be applied to reduce random variations in pixel intensities while preserving important structures like GGO boundaries. Since CT images inherently contain quantum noise, it can create false GGO detections.

4.3. Lung Segmentation

In order to perform Lung segmentation, we will use a combination of intensity based thresholding and morphological image processing techniques. First we will find the real area of the lung in mm^2 by finding the real size of the pixel dimensions. Nex, we will binarize the images using intensity thresholding to separate lung tissues from the background. We expect lungs to be in the Housendfield unit range of [-1000,-300]. To this end, we need to clip the image range to [-1000,-300] and binarize the values to 0 and 1.

Then we need to find the contours of the lung areas. In computer vision, a contour is a set of points that describe a line or area. So for each detected contour we will not get a full binary mask but rather a set with a multiple x and y values. In order to isolate the desired area we will use the marching squares algorithm[]. Using the possible contours, we will extract the lungs using 2 constraints:

- The contour of the lungs must be a closed set
- The contour must have a minimum volume of 2000 pixels to represent the lungs

Using these 2 assumptions, we will convert the contours into a binary mask representing the lung areas by using the ”pillow” Python library to draw a polygon connecting the contours After obtaining the binary mask of the lungs, we will save it as a NifTI file and use it for further processing. Finally, we will do element-wise multiplication between the CT image and the lung mask we obtained to get only the lungs.

4.4. GGO Visualization and Evaluation

After segmenting the lungs, we will focus on visualizing the GGO regions within the lung areas. We will assume that if there is a pixel with an intensity value between -500 HU and -700 HU inside the lung area then we will consider it as a GGO area. However, this threshold range may also include vessels within the lung parenchyma. To address this issue, we will apply vessel removal techniques to reduce false positive detections. Since vessels appear as elongated, tubular structures, we will use morphological opening operations using a circular structuring element to remove them. Then, we will set the zeros that resulted from the previous element-wise multiplication to -1000 and finally keep only the intensities that are between -500 and -700 as GGO areas.

Once the GGO area has been identified, we will visualize it by overlaying the detected GGO regions on the original CT images using pseudo-coloring techniques. We will also visualize the original image alongside the segmented GGO area side-by-side using a User Interface. Finally, we will evaluate the effectiveness of the GGO segmentation using manually annotated contours provided in the XML files of the LIDC-IDRI dataset. Back to Table of Contents

5. Dataset / Data Collection

This project utilizes the LIDC-IDRI dataset obtained from The Cancer Imaging Archive (TCIA). The dataset consists of over 1,000 chest CT scans stored in DICOM format. The key statistics of the dataset are summarized as follows:

- **Total cases:** 1,018 patients
- **Total lesions:** 7,371 lesions marked as "nodule" by at least one radiologist
- **Nodules ≥ 3 mm:** 2,669 lesions identified as clinically significant nodules

Images: Clinical thoracic CT scans in DICOM format.
Annotations: Provided in XML files containing the coordinates of nodule boundaries (contours) and associated radiological characteristic ratings.
Tools: The Python library `pylidc` is used to query, process, and visualize the LIDC-IDRI dataset.

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6. Expected Outcomes

The expected outcomes of this project include:

- A Python based image processing pipeline for lung CT analysis
- A GUI application for loading, visualizing, and navigating CT slices
- Accurate lung segmentation masks
- Highlighted Ground Glass Opacity (GGO) regions with color overlays
- Quantitative outputs such as GGO count, area, and estimated volume
- Adjustable HU threshold controls for experimentation
- Side by side visualization of original and processed images

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7. Percentage of Image Processing & Computer Vision Involvement

Core components include DICOM loading, Hounsfield Unit extraction, lung windowing, threshold-based lung segmentation, morphological operations, connected component analysis, flood fill, and GGO detection using HU ranges. Vessel suppression is performed using structural morphology and region-based filtering. Dataset preprocessing and visualization parts of the pipeline are the only components that do not involve image processing or computer vision techniques.

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